FINAL PROJECT REPORT

EV Market Performance & Insights

Data Analytics with Tableau

Project Type	Individual Project
Tool	Tableau Public Desktop
Publication	Tableau Public Story
Report Date	31 October 2026

Prepared as an individual academic project report. This document follows the section order in the supplied Final Report Template.

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1. INTRODUCTION

1.1 Project Overview

EV Market Performance & Insights is a Tableau-based data analytics project that explores electric-vehicle market information through key performance indicators, comparisons and visual storytelling. The dashboard is designed to help a viewer examine sales patterns across years, countries, manufacturers, vehicle categories and battery technologies, alongside driving range, customer rating and vehicle price.

The project uses a CSV data source connected directly to Tableau Public Desktop. The completed workbook includes KPI cards, charts, a geographic map, a dashboard and a Tableau Story. It is an exploratory analytics dashboard; it is not a live transaction system, predictive model or Snowflake implementation.

1.2 Purpose

The purpose is to convert tabular EV market data into understandable visual insights. The dashboard supports quick review of overall performance and helps users compare market segments without manually scanning rows of data.

Project objectives

• Summarise total EV sales, average driving range and average customer rating.

• Show the year-wise EV sales trend.

• Compare sales by vehicle category, battery technology, country and manufacturer.

• Show the geographic distribution of sales.

• Explore the relationship between average vehicle price and customer rating.

• Present the key observations through a Tableau Story.

2. IDEATION PHASE

2.1 Problem Statement

EV market data contains multiple dimensions such as year, country, manufacturer, vehicle category and battery technology, as well as measures such as sales, driving range, customer rating and price. Reviewing these fields separately can make it difficult to identify trends and compare segments. The project addresses this by bringing the available information into a consolidated, visual Tableau dashboard.

2.2 Empathy Map Canvas

The intended dashboard user is an analyst, student or business reviewer who wants to understand the EV market quickly. The following canvas is a project-level user perspective, not the result of a formal user interview.

Thinks	**Feels**
Which years and markets show higher sales? Which vehicle and battery segments are more prominent? How do price and customer ratings compare?	Needs a clear, trustworthy summary and may feel overwhelmed by a large raw dataset.
Says	**Does**
“I want to compare EV market performance in one place.”	Uses KPI cards, filters, charts, map and Story points to explore the available data.

2.3 Brainstorming

• Use KPI cards for a quick market overview.

• Use a line chart to show sales change over time.

• Use bar charts to compare countries, vehicle categories and battery technologies.

• Use a pie chart to compare manufacturer contribution.

• Use a map to show geographic sales distribution.

• Use a scatter plot to compare average vehicle price and customer rating.

• Arrange the views into a dashboard and create a Story to communicate the findings.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User action	Need / opportunity
Discover	Opens the Tableau dashboard.	Understand the project topic and overall market KPIs.
Explore	Uses the Year filter and reviews charts.	Compare years, countries and market segments.
Interpret	Reads tooltips, legends and Story captions.	Understand patterns and identify areas for further analysis.

3.2 Solution Requirement

• Functional: connect the supplied CSV file to Tableau.

• Functional: display total sales, average driving range and average customer rating.

• Functional: provide year-wise and segment-wise sales comparisons.

• Functional: provide geographic and manufacturer views.

• Functional: provide a dashboard and explanatory Story.

• Non-functional: readable labels, meaningful chart titles, consistent formatting and a clear layout.

• Non-functional: calculations and observations should be validated against the connected source data.

3.3 Data Flow Diagram

Step	Description
1. Data source	Provided EV market CSV file.
2. Data connection	CSV is connected in Tableau Public Desktop.
3. Analysis	Dimensions and measures are used to aggregate sales, range, rating and price.
4. Visualisation	KPI cards, charts and map are created as worksheets.
5. Presentation	Worksheets are arranged in a dashboard and Tableau Story.

3.4 Technology Stack

Component	Technology / use
Data source	CSV dataset supplied for the project
Analytics and visualisation	Tableau Public Desktop
Publishing	Tableau Public
Documentation and sharing	PDF/Word documents; Google Drive and GitHub planned for submission packaging

4. PROJECT DESIGN

4.1 Problem Solution Fit

A consolidated Tableau dashboard is suitable for the stated problem because it presents the key measures and market comparisons in one place. KPI cards support quick review, while trend, comparison, map and scatter views answer

different analytical questions. The Story provides a guided sequence for interpreting the views.

4.2 Proposed Solution

Connect the EV CSV file to Tableau, review fields and data types, create worksheets for the required KPIs and market comparisons, assemble the worksheets into a dashboard, add available filters, and create a Tableau Story to communicate the findings.

4.3 Solution Architecture

Layer	Role
CSV data	Source records for year, country, manufacturer, category, battery technology, sales, range, rating and price.
Tableau connection	Loads the CSV fields into Tableau as dimensions and measures.
Analysis	Uses SUM and AVG aggregations to summarise sales, range, rating and price.
Presentation	Displays KPI cards, charts, map, dashboard and Story.
Reviewer / user	Explores the dashboard and interprets the visual insights.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was organised into sequential activities: dataset access and preparation, KPI and chart creation, dashboard and Story development, validation, documentation, and final publication/submission. The planning template contained four proposed sprints and user stories. The sprint schedule should be treated as a plan, not proof that each activity was completed on those exact dates.

Workstream	Planned output
Data access & preparation	Connected CSV and reviewed fields.
Analysis & visualisation	KPI cards and market comparison worksheets.
Dashboard & Story	Consolidated dashboard, filter and story points.
Validation & publishing	Check calculations and publish Tableau Public workbook.
Documentation & submission	Final report, evidence, demo video and repository packaging.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The Tableau performance template records the following visible project features: a CSV source connected to Tableau; a Year filter shown on the dashboard; aggregation measures for sales, driving range, customer rating and price; seven chart views and three KPI cards; and five visible Story-point captions.

The available screenshots do not provide measured data-load time, filter response time, query execution history or a controlled performance comparison. These values are therefore not reported as measured results. The exact CSV row count, file size and preprocessing audit should be verified from the final workbook's connected data source before submission.

Functional checks recommended before final submission

- Open the published Tableau link and confirm that the dashboard and Story load.
- Change the Year filter and verify that the intended worksheets respond.
- Check that KPI cards and chart labels are readable.
- Compare displayed totals and averages with the exact connected CSV source.

• Check that map marks, pie chart legend and scatter plot marks are visible.

7. RESULTS

7.1 Output Screenshots

The following figures are taken from the Tableau screenshots supplied during project development. The values are transcribed from the screenshots and should be reconciled with the final connected data source before formal evaluation.

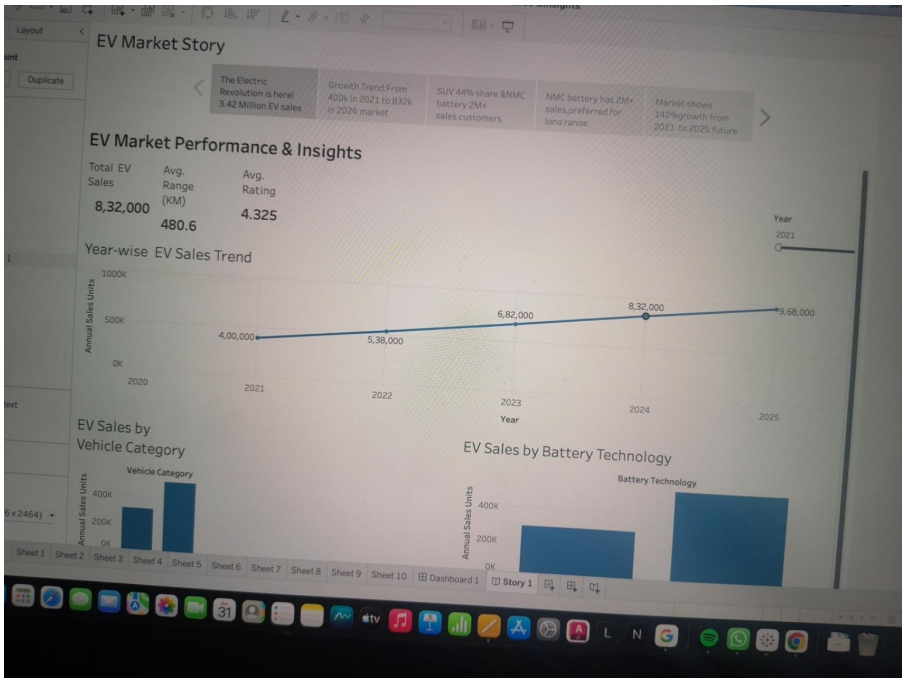


Figure 1. Tableau EV Market Story overview and KPI cards.

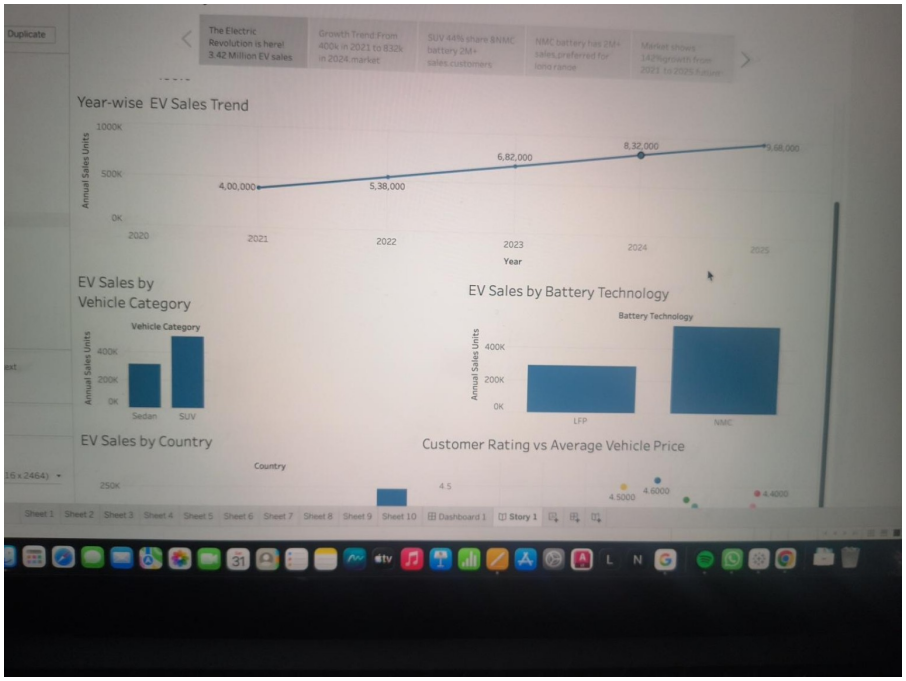


Figure 2. Year-wise EV sales trend and category/battery technology charts.

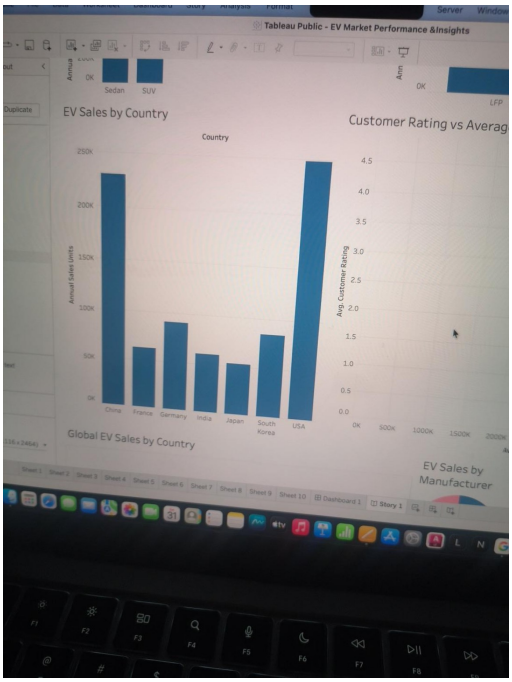


Figure 3. Country comparison and customer-rating versus price view.

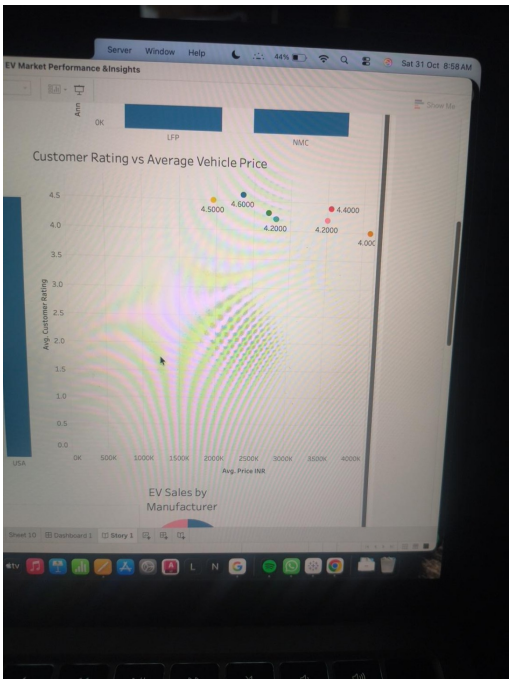


Figure 4. Customer rating versus average vehicle price scatter plot.

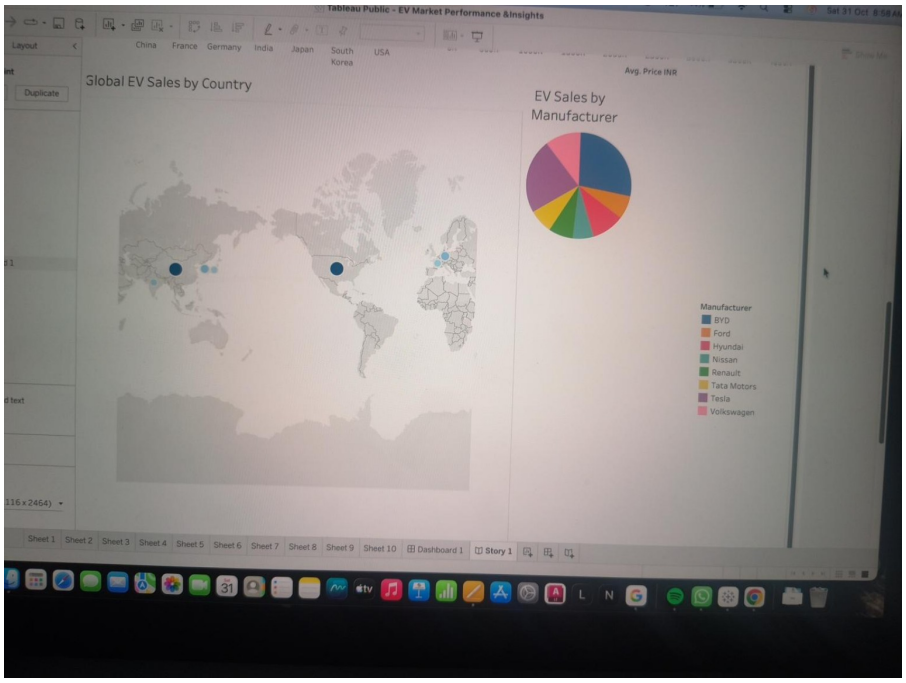


Figure 5. Global EV sales map and manufacturer contribution pie chart.

7.2 Key Observations

• The dashboard screenshot displays KPI values of 832,000 total EV sales, 480.6 km average driving range and 4.325 average customer rating.

• The year-wise line chart labels sales increasing from 400,000 in 2021 to 968,000 in 2025.

• The visible vehicle-category comparison shows SUV sales higher than Sedan sales.

• The battery technology chart shows NMC sales higher than LFP sales.

• The country chart compares China, France, Germany, India, Japan, South Korea and the USA.

• The map shows geographic distribution across the represented countries.

• The manufacturer pie chart includes BYD, Ford, Hyundai, Nissan, Renault, Tata Motors, Tesla and Volkswagen.

• The scatter plot compares average price and average customer rating across manufacturer marks.

8. ADVANTAGES & DISADVANTAGES

Advantages

• Combines several EV market measures in one dashboard.

• Provides a quick summary through KPI cards.

• Supports year-wise, country-wise and segment-wise comparisons.

• Uses a geographic map and scatter plot to provide different analytical perspectives.

• The Tableau Story helps communicate findings in a guided sequence.

Disadvantages / limitations

• The analysis is limited to the fields, years and records available in the supplied dataset.

• The dashboard is based on a static CSV source unless the workbook is refreshed with updated data.

• The supplied screenshots show a discrepancy between the Story caption (3.42 million sales) and the dashboard KPI (832,000 sales); the source must be reconciled.

• No Snowflake implementation, predictive model or measured query-performance benchmark is included.

9. CONCLUSION

The EV Market Performance & Insights project demonstrates how Tableau can be used to convert EV market data into an interactive visual analysis. The dashboard brings together KPI cards, sales trends, category and battery comparisons, country-level views, a map, manufacturer contribution and customer-rating versus price analysis. The Tableau Story provides a guided summary of the observations. Before final submission, the connected dataset and the figures shown in the dashboard and Story should be reconciled to ensure consistent reporting.

10. FUTURE SCOPE

• Connect a refreshed or regularly updated EV dataset.

• Add more detailed filters such as country, manufacturer, vehicle category and battery technology if required.

• Add validated year-over-year growth and market-share calculations.

• Introduce forecasting only if a suitable time series and a validated forecasting method are available.

• Improve dashboard layout for different screen sizes and reduce unnecessary scrolling.

• Add a documented data-quality audit and measured dashboard performance tests.

11. APPENDIX

11.1 Source Code (if any)

No standalone source-code application was developed. The main implementation is a Tableau workbook containing worksheets, dashboard and Story. Tableau aggregation expressions documented for the KPI and chart views include SUM([Annual Sales Units]), AVG([Driving Range KM]), AVG([Customer Rating]) and AVG([Price INR]).

11.2 Dataset Link

The dataset was supplied through the project workspace. A public dataset URL was not provided in the materials available for this report. Add the approved dataset source or Drive link here if required by the reviewer.

11.3 GitHub & Project Demo Link

Tableau Public project link: Open EV Market Performance & Insights – Tableau Public

GitHub repository: To be added after the repository is created and made accessible.

Project demonstration video: To be added after the screen-recorded walkthrough is uploaded to Google Drive.

Final submission checklist

• Verify the connected CSV source and reconcile all KPI and Story figures.

• Confirm the Tableau Public link opens the final dashboard and Story.

• Add the final dataset link if the dataset is shareable.

• Create the GitHub repository and add the Tableau workbook, templates, documentation and demo video.

• Replace the pending GitHub and demo references in this report with their final URLs.

PROJECT ACCESS & SUBMISSION LINKS

EV Market Performance & Insights — use the links below to access the published dashboard, project demonstration recording and source repository.

1. Interactive Tableau Public Dashboard

The published Tableau Story contains the EV Market Performance & Insights dashboard and its story points.

Open EV Market Performance & Insights on Tableau Public

https://public.tableau.com/app/profile/joha.khan/viz/EVMarketPerformanceInsights_17930988283380/Story1?publish=yes

2. Project Demonstration Video (Voice-over)

The demonstration recording presents the dashboard walkthrough with voice-over narration.

Watch the Project Demonstration Video on Google Drive

https://drive.google.com/file/d/1fRQ_uXoGiN5zEriWVdtisahefWb5g7Ov/view?usp=sharing

3. GitHub Repository

The repository contains the project README and uploaded project documentation/resources.

Open the EV Market Performance & Insights GitHub Repository

https://github.com/johak22/EV-Market-Performance-Insights

Submission note: Ensure the Google Drive video is shared with “Anyone with the link” permission set to Viewer, and keep the Tableau Public workbook published and accessible.